Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent, suspension

Species Used Human, K562, chronic myeloid leukemia; Hamster, CHO, ovary; Hybrid, rat / mouse, MEL cells.

Molecules Electroporated DNA: linearized constructs

Before the Pulse

Cell Growth Medium RPMI 1640, 10% Fetal Bovine Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest 3 x 10 (5) to 5 x 10 (5) cells / ml (suspension)

Pre-pulse Incubation 10 min. at room temp in Hepes buffered saline with dextrose

Wash Solution Phosphate Buffred Saline

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature Room temperature

Electroporation Medium* HEPES buffered saline with dextrose

Cell Density 2 x 10 (7) cells / ml

Volume of Cells 0.4 ml

DNA Concentration Not given

DNA Resuspension Buffer RPMI 1640 +10% Fetal Bovine Serum

Volume of DNA Not given

Cuvette Gap 0.4 cm

Voltage 0.2 kV

Field Strength 0.5 kV/cm

Capacitor 960 µF

Resistor (Pulse Controller) Ω none

Time Constant 25.0 msec

After the Pulse

Outgrowth Medium RPMI 1640+10% Fetal Bovine Serum

Outgrowth Temperature 37 °C

Length of Incubation Not given

Selection Method or Assay Used G418

Electroporation Efficiency Not given

Per Cent Survival Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH_2PO_4, 1.15g Na_2HPO_4
HBS: 10mM HEPES,pH 7.2,150 mM NaCl, 5 mM $CaCl_2$

Name of Submittor No name or address given

Institution Address

Survey Number 168